

Public Safety Analytics and Database Management System

**A practical platform for turning public campus safety records
into clear, usable, and repeatable insights**

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Data coverage: 2023–2025 public incident and arrest records

1. Executive Summary

Terp Protect was created to make public campus safety information easier to understand, compare, and use. Public incident and arrest records contain valuable information, but they are spread across separate files, use inconsistent labels, and require extensive manual effort before meaningful patterns can be identified.

The project brings these records into one organized platform, applies consistent definitions, connects related incidents and arrests, highlights data-quality issues, and presents the results through an interactive dashboard. The outcome is a clearer view of when incidents occur, where activity is concentrated, how cases are resolved, how quickly incidents are reported, and how arrest records relate to incident records.

The project's core value is simple: it converts fragmented public records into a reliable, repeatable, and decision-ready source of information.

2. Project Motivation and Business Problem

Campus safety information is often available to the public, but availability alone does not make it useful. Reviewing multiple files manually is slow, definitions may vary from year to year, and separate incident and arrest records are difficult to compare.

The project was motivated by the need to answer practical questions more efficiently:

- Which types of incidents occur most often?
- Where is public-safety activity concentrated?
- When do incident volumes increase?
- How quickly are incidents reported?
- What outcomes are most common?
- How many incident records can be connected to arrest records?
- Which records require additional review before analysis?

Without a unified system, these questions require repeated manual work. Terp Protect addresses that gap by creating one consistent source for recurring analysis.

3. Goals and Objectives

The project was designed around five practical goals:

- Create a single, organized view of public incident and arrest records.
- Standardize inconsistent crime, location, outcome, and charge descriptions.
- Reduce repetitive manual preparation through an automated workflow.
- Make important trends easy to explore through a clear dashboard.
- Preserve transparency by identifying records that may require review.

Success is measured by whether the platform can reproduce the analysis consistently, explain its calculations clearly, and help users move from raw records to useful findings with less effort.

4. Target Users and Their Needs

User Group	Primary Need
Public-safety teams	Need a consolidated view of incident patterns, outcomes, reporting timeliness, and arrest linkages.
University administrators	Need clear summaries that support planning, resource discussions, and operational awareness.
Analysts and researchers	Need consistent definitions, reusable data, and transparent calculation rules.
Students and the campus community	Need public information presented in a form that is easier to interpret.
Project reviewers and collaborators	Need a clear understanding of the project purpose, logic, limitations, and future potential.

5. Proposed Solution

Terp Protect provides a simple end-to-end flow: collect the records, clean and standardize them, connect related records, validate important fields, and present the results in an interactive dashboard. The platform does not replace official public-safety systems. Instead, it adds a practical analytical layer that makes existing public information easier to review and reuse.

How the information is organized

Incident dates, crime descriptions, locations, case outcomes, arrest charges, and related identifiers are organized using consistent categories. This allows the same definitions to be used across years and across dashboard sections.

How users interact with the solution

Users can filter the records and review summary measures, trend charts, category comparisons, reporting delays, arrest linkages, and data-quality checks without writing code or manually combining files.

6. Dashboard Overview

The dashboard is organized into seven focused views. Each view answers a different business question while using the same underlying records and definitions.



Command Center

Provides a high-level view of incident volume, leading categories, case outcomes, reporting timeliness, and overall coverage.

Time and Seasonality

Shows how incident activity changes by month, academic period, weekday, and hour.

Location Analysis

Identifies high-activity locations and compares the types of incidents recorded across location groups.

Incident Outcomes

Explains how cases are categorized as closed, pending, arrest-related, or assigned to other outcomes.

Reporting Delay

Measures how quickly incidents are reported and highlights categories with longer delays.

Arrest Analysis

Summarizes arrest volume, major charge categories, and links between incident and arrest records.

Data Quality

Shows whether important fields are complete and identifies records that require review.

7. Key Findings

The current analysis covers 5,659 incident records and 1,175 cleaned arrest records from 2023 through 2025. Key findings include:

- Medical / Welfare is the largest standardized incident category, showing that campus public-safety activity extends beyond traditional crime response.
- Roadway / Street is the leading location group by incident volume, indicating a strong concentration of activity in transportation and public-access areas.
- Approximately 49.4% of incidents are categorized as closed or cleared, while about 29.8% remain pending or active.
- Approximately 81.8% of incidents with valid delay information were reported within 24 hours.
- The median reporting delay is approximately 0.2 hours, while the 90th-percentile delay is approximately 3.9 days.
- Approximately 17.3% of incidents are linked to at least one arrest record.
- DUI / Impaired Driving is the largest standardized arrest charge category.
- The primary field-check pass rate is approximately 99.94%, with a small number of incident records retained for review.

These findings provide a clear baseline for understanding activity patterns, reporting behavior, case progress, and record quality. They should be interpreted as descriptive observations rather than measures of individual or location-based risk.

8. Business and Operational Value

Terp Protect creates value by reducing the effort required to move from raw records to usable information. The platform can support:

- Faster recurring analysis by reducing manual file preparation.
- More consistent reporting through shared definitions and calculations.
- Improved visibility into patterns that may otherwise remain hidden across separate files.

- Clearer communication through simple, interactive views.
- Earlier identification of incomplete or invalid records.
- A reusable foundation that can be extended as new records become available.

The most important impact is not a single chart or metric. It is the creation of a dependable process that can be repeated, reviewed, and improved over time.

9. Key Challenges and Decisions

Inconsistent source terminology

Similar incidents and locations appeared under different labels. The project introduced broader standardized categories while retaining source-level detail.

Separate incident and arrest records

The records were published independently. Case numbers were cleaned and used to connect related records where a reliable match was available.

Invalid reporting sequences

A small number of records showed reporting timestamps earlier than occurrence timestamps. These records were retained for transparency but excluded from delay calculations.

Avoiding misleading comparisons

Raw counts can be mistaken for risk rates. The dashboard therefore uses clear definitions, minimum sample thresholds, and interpretation notes.

Balancing detail and usability

Too much technical or analytical detail can overwhelm users. The dashboard was organized into focused sections with compact explanations and review panels.

10. Limitations and Responsible Use

The project uses publicly available records and does not represent the complete information available within internal public-safety systems. Its findings are descriptive and should not be treated as causal conclusions.

- Incident counts are not population-adjusted risk rates.
- High-volume locations may also have greater traffic, exposure, operating hours, or visibility.
- Record matching depends on available and consistent case numbers.
- Standardized categories support analysis but do not replace official terminology.
- Academic periods use simplified calendar groupings.
- The platform should not be used for individual profiling, risk scoring, or automated enforcement decisions.

The solution is best used for reporting, trend review, data validation, process improvement, and identifying questions that require deeper investigation.

11. Future Improvements

- Add scheduled data refreshes for new public records.
- Improve location standardization and introduce map-based analysis.
- Use exact academic-calendar dates instead of simplified periods.

- Add automated post-pipeline validation for files, schemas, row counts, and database objects.
- Support longer historical coverage and additional public datasets.

12. Conclusion

Terp Protect demonstrates how public records can be transformed into a clear and useful operational resource. By organizing fragmented information, applying consistent definitions, connecting related records, validating important fields, and presenting findings through an accessible dashboard, the project creates a stronger foundation for recurring campus safety analysis.

Its value lies in making important information easier to find, easier to explain, and easier to use. The platform establishes a practical base for future expansion while maintaining transparency, responsible interpretation, and respect for the limits of public data.